



Figure 3: Screen magnifiers

cs.washington.edu/wa.php. As soon as this Web page gets rendered, your screen reader is ready. You can type any URL and start accessing it in an audio format. WebAnyWhere has been developed by the WebInSight group of the University of Washington. The other accessibility projects by the same group can be viewed at <http://websight.cs.washington.edu/projects/>.

- **TalkBack:** As the use of smartphones increases tremendously, the need for screen readers for them is inevitable. TalkBack is Android's accessibility software for the visually challenged. It becomes active when the user turns on *Accessibility*. In versions 3.2 and earlier, to turn it on, go to *Settings* and click on *accessibility*. Then select *check box*, and click on *TalkBack Check box*. In versions 4.1 and later, enable the TalkBack switch.

Screen magnifiers

In the case of low-vision users and the elderly, who can't see the text or images of a smaller size, screen magnifiers can help them carry out their tasks in a smoother and more efficient manner.

- **ORCA:** In the case of the Linux environment, ORCA performs screen magnification in addition to screen-reading.
- **Virtual Magnifying Glass:** This is a cross-platform, free and open source magnifier. Virtual Magnifying Glass is an efficient magnifying tool. It can be downloaded from <http://magnifier.sourceforge.net/>. It supports multiple monitors.
- **MagMe:** The screen magnifier that can run from a USB drive is MagMe. It is free and open source and targeted for the Windows OS. MagMe is developed by the *Computer Aid Adaptive Software 4BVI Project* and is designed to work in parallel with NVDA. It can be downloaded from <http://sourceforge.net/projects/magme/>.

In addition to these screen readers and screen magnifiers, there is an array of hardware support like Braille displays, etc, for the visually challenged.

Figure 4: Screenshot of the OSFY's website www.opensourceforu.com with the open dyslexic Chrome extension enabled

For the hearing impaired

The accessibility features required for those with hearing problems are completely different from the requirements of the visually challenged. The former need tools like speech-to-text converters, sub-titles for audio clips and other visual alerts.

To convert speech into text, there are various speech recognition engines. One such powerful engine is CMU Sphinx, developed by Carnegie Mellon University. The features included in this tool kit are speech-to-text, audio transcriptions etc. CMU Sphinx can be downloaded from <http://sourceforge.net/projects/cmusphinx/>.

An assistive tool that would be very useful for auditory challenged people is *Hangout Captions*. Those with hearing difficulties can use this tool to convert speech into text in real time during Google Hangout sessions. For further information on this tool, visit <https://hangout-captions.appspot.com/>.

There are various specialised gadgets for auditory challenged people. Interested users can go to <http://www.deafgadgets.com/> to keep themselves updated with recent happenings in this domain.

For the cognitively challenged

There are some specific efforts to make computers accessible to people with cognitive challenges. Let us focus on an important and widely prevalent challenge—dyslexia. Wikipedia defines dyslexia or the developmental learning disorder as difficulty with learning to read fluently and with accurate comprehension despite normal intelligence.

Dyslexic people would find it difficult to read the text appearing on the screen. There are specialised fonts that make reading simpler for them. One such open source effort is at <http://opendyslexic.org/> which provides a free and open source typeface. By simply installing the font or browser extension, you would get this feature in your system. If you are already using Linux distros like Fedora, Arch Linux, etc, this package is available in their default software repositories.

A screenshot of the *Open Source For You* home page